

Network Selection Problem for Voltage Regulation Device Installation

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Edinburgh Undergraduate Operational Research Challenge – October 2025

1 Introduction

This challenge problem lies in the domain of power system modelling. The company seeks to select a subset of distribution networks in which to install a voltage regulation device at the substation of each network. The device regulates the substation output voltage and can be regarded as a step-down transformer (from medium-voltage primary to low-voltage secondary) equipped with voltage regulation. For background on power system modelling, see [1], in particular “Chapter 2 Fundamentals”.

An initial screening step has already produced a pool of candidate networks for which the device is compatible. In technical terms, this means that the device’s ratings are compatible with the network (e.g., primary/secondary rated voltages match nominal levels and the expected maximum loading does not exceed the device’s power rating). The task is to choose a set of networks to install the devices from this candidate set.

The device can maintain the substation’s low-voltage (secondary) bus within a specified regulation band (e.g., $\pm 5\%$ around the nominal voltage of 230 V). However, ensuring acceptable operation requires that *all* customer node voltages along the network remain within statutory limits (typically $\pm 10\%$ of nominal voltage) throughout the year. Fixing the secondary bus does not guarantee compliance at downstream nodes, whose voltages depend on loading, generation, and line impedances.

Node voltages are driven by (i) net demand or generation at each connection point (including prosumers with rooftop solar panels who are net contributors to the grid) and (ii) the distribution network’s topology and impedances. Heavily loaded networks tend to experience voltage drops that can cause undervoltage, while lightly loaded feeders with significant local generation may experience voltage rise and possible overvoltage. Higher-resistance conductors and longer, more spread-out feeders exhibit greater voltage drops (and losses).

In this challenge, only high-level data are available: numbers of customers/prosumers, average consumption/generation, load profiles for summer and winter, and a simplified topology for the networks. To enable tractable modelling and verification, we assume a one-dimensional serial network topology, as depicted in Figure 1 below, with typical line parameters (resistance, reactance, etc.). Using these inputs, the aim is to identify those candidate networks for which the device’s regulation capability is sufficient to keep all node voltages within operational limits across representative operating conditions.

2 Problem Statement

In real-world systems, electrical quantities such as power, voltage, and current are generally complex values ($\in \mathbb{C}$), and distribution may be three-phase rather than single-phase. Here, we adopt a simplified model: all quantities are taken to be real-valued ($\in \mathbb{R}$), networks are treated as single phase, and the lines between nodes are assumed to be purely resistive (no inductance or capacitance), thus avoiding imaginary terms. For line modelling fundamentals, see Chapters 4–5 in [1].

The task should be approached in two stages. In Stage 1, you will use the simplified assumptions described above to build and analyse the model. In Stage 2, you will progressively relax these simplifications to assess the robustness of the conclusions. This two-stage process will highlight which parameters most strongly influence the final network selection decisions and will help quantify how much confidence can be placed in results derived from the simplified model, given uncertainty in parameter estimates.

An important factor in distribution networks is topology, which is normally determined from geographical data on line locations. For high-voltage transmission systems, such data are relatively accessible (e.g., through tools such as <https://openinframap.org/>), but at the low-voltage level it is rarely public and must often be compiled manually (e.g., by visually tracing line routes in satellite imagery). For the purposes of Stage 1, you will assume that each network is *serial*: all nodes are connected in a single chain from the substation to the furthest downstream node, as if it were laid out on one straight street.

In practice, load and generation at customer nodes vary over time, causing voltages to fluctuate. For instance, during summer months, photovoltaic generation i.e. (solar panels) may dominate and demand may be low, leading to potential voltage rise. In winter, heavy heating loads may cause significant voltage drops. A voltage regulation strategy should adapt to such seasonal variations—for example, by setting the secondary bus slightly above nominal during winter to mitigate undervoltage. For this exercise, you may assume that the regulation set-point can be chosen optimally for each profile, as though the load and generation patterns were known in advance.

Two representative profiles are provided: *peak winter* and *peak summer*, representing worst-case loading scenarios. Because installation of the voltage regulation device is a long-term investment, the selected networks must remain compliant under all such worst-case conditions. The objective is to verify that the device’s regulation range is sufficient to maintain all node voltages within statutory limits in both profiles.

2.1 Task

Your task is to develop a tractable mathematical model using the information provided, and to apply this model to recommend a suitable subset of networks for installation of the voltage regulation device. Beyond producing a selection, you should examine the limitations of your modelling approach and assess the robustness of the recommended networks to uncertainty in key parameters (e.g., through sensitivity analysis and risk quantification).

The aim is not to find a single “correct” answer, but to provide a reasoned, evidence-based selection and to highlight how modelling assumptions and incomplete data influence the outcome. The emphasis is on improving understanding of how such network selection decisions can be made when only partial information is available, and on providing the company with insights to support more informed long-term investment choices.

2.2 Data

You are provided with a CSV file containing specifications for 20 000 candidate networks. From these, you must identify 10 suitable networks for installation of the voltage regulation device.

2.2.1 Fixed Parameters

The following parameters are identical for all networks:

- **Nominal voltage:** 230 V — secondary-side voltage under baseline (no regulation) conditions.
- **Minimum acceptable voltage:** 207 V — statutory lower bound for healthy operation at any node.
- **Maximum acceptable voltage:** 253 V — statutory upper bound for healthy operation at any node.
- **Voltage regulation increase limit:** +5% — maximum upward adjustment relative to the *current* secondary-side voltage.
- **Voltage regulation decrease limit:** −5% — maximum downward adjustment relative to the *current* secondary-side voltage.

2.2.2 Network-Specific Data

Each row in the CSV file corresponds to a single network and contains the following fields:

- **Size N :** Total number of nodes in the network, including the substation's secondary side (p_0).
- **Line resistance R (Ω/km):** Per-unit-length resistance, assumed uniform for all lines within the network.
- **Line length L (km):** Physical length of each line segment between two consecutive nodes, assumed identical for all such segments in the network.
- **Winter power profile P (W):** Power injections per node for the peak winter period, given as $p_0, p_1, \dots, p_N$ and separated by commas.¹
- **Summer power profile P (W):** Power injections per node for the peak summer period, also given as $p_0, p_1, \dots, p_N$ and separated by commas.

2.3 Objective Functions

There are two primary objectives in this project. The first is to identify the largest networks in which the voltage regulation device can maintain healthy operation (i.e., all node voltages within statutory limits) in both the peak summer and peak winter profiles.

The second objective is to minimise the regulation effort. This could be quantified, for instance, by the average or maximum absolute voltage adjustment needed across all operating scenarios. Such a multi-objective approach can help prioritise networks that not only meet the voltage constraints but also provide greater operational margin and resilience.

2.4 Voltage Model

To model the voltages across the nodes in the network, given the voltage set at the secondary side of the transformer and the power values at the nodes, you will need to derive the currents across the lines. Normally, you would construct an admittance matrix, which is often called bus or nodal admittance matrix (see [2]), that depends on the connectivity of the network. This matrix builds the relationship between the currents and voltages at every node. You are free to use a model using this admittance matrix as described in [2]². However, it is OK to use simpler models. Below, we provide a simplified model where voltages and currents are modelled in a sequential basis, as if disregarding interdependence, which would not be too bad to start with since we are assuming loosely connected serial networks.

In Figure 1, we illustrate the voltage model in a general serial network (for a fixed power profile). To know how the voltage will drop/increase as it traverses a line and reaches a node with a given power value, you would need a model (this is the function f in the illustration). You can assume the approximate voltage drop model in [3] to infer the voltage drop across a line i by ZI , where Z denotes the total impedance over the line computed as $Z = LR$ (L : total line length, R : line resistance) and $I = P_{i+1}/V_{i+1}$, based on the receiving end of the line. You can also consult the section on short lines in Chapter 5 of [1] to see the theory on the relationship between the voltages and currents at the sending (V_s, I_s) and receiving (V_r, I_r) ends (see Figures 2-3).

Note that you are free to choose any reasonable voltage model. This model above is only presented for showing an example. There are alternatives [4].

2.5 Robustness Investigation

Stage 2 of the challenge focuses on evaluating the robustness of the network selection obtained from the simplified Stage 1 model. It is acceptable to base the initial selection on a simplified representation (e.g., basic voltage model, serial network topology, and two seasonal load profiles) provided that the robustness of this selection is assessed and justified.

¹ p_0 represents the secondary bus and is always zero; negative values indicate consumption, positive values indicate generation.

²Admittance terms are simply the inverse of impedance.

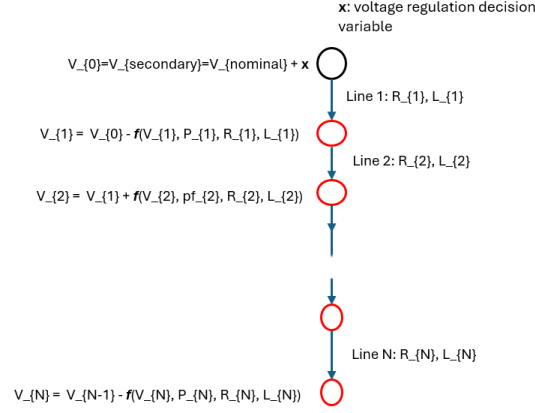


Figure 1: Illustrating Serial Network Voltage Model

$$\begin{bmatrix} V_S \\ I_S \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_R \\ I_R \end{bmatrix}$$

Figure 2: Equation 5.1.3 in [1]

Parameter	A = D	B	C
Units	per Unit	Ω	S
Short line (less than 80 km)	1	Z	0
Medium line—nominal π circuit (80 to 250 km)	$1 + \frac{YZ}{2}$	Z	$Y \left(1 + \frac{YZ}{4}\right)$
Long line—equivalent π circuit (more than 250 km)	$\cosh(\gamma\ell) = 1 + \frac{Y'Z'}{2}$	$Z_c \sinh(\gamma\ell) = Z'$	$(1/Z_c) \sinh(\gamma\ell) = Y' \left(1 + \frac{Y'Z'}{4}\right)$
Lossless line (R = G = 0)	$\cos(\beta\ell)$	$jZ_c \sin(\beta\ell)$	$\frac{j \sin(\beta\ell)}{Z_c}$

Figure 3: Table 5.1 in [1]

This stage involves systematically exploring the sensitivity of results to variations in key problem dimensions, such as network topology, line parameters, and power profiles so that risk-related questions can be addressed. Examples include:

- *Load uncertainty:* If the actual node consumption were, for example, 10% higher than expected (due to future demand growth), would the device still be able to maintain healthy operation?
- *Topology uncertainty:* If the actual network layout were radial or branched rather than the assumed serial structure, how likely is it that the device could still meet voltage constraints?

The depth and breadth of this investigation are at your discretion; any well-founded insights will be valuable to the company. Possible avenues include:

- Evaluating alternative network topologies beyond the serial assumption.
- Employing more detailed voltage models that account for additional electrical effects.
- Introducing perturbations or errors into the power profiles to represent forecasting inaccuracies or behavioural changes.

The aim is to quantify confidence in the selection decisions and to identify the factors that most critically affect device performance under realistic uncertainties.

References

- [1] J. Duncan Glover, Mulukutla S. Sarma, Thomas J. Overbye *Power System Analysis and Design*, 5th Edition, CENGAGE Learning. 2012.
- [2] https://home.engineering.iastate.edu/~jdm/ee458_2011/PowerFlowEquations.pdf

- [3] <https://wzy.ece.iastate.edu/Courses/EE455/03%20EE455%20Approximate%20Method%20of%20Analysis.pdf>
- [4] <https://engineering.purdue.edu/~kekatos/pdsa/Lecture11.pdf>